

Understanding Reconstruction-Based Visual Token Compression: When and Why Does ApET Work?

1. Project Title

Understanding Reconstruction-Based Visual Token Compression: When and Why Does ApET Work?

2. Group Members

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3. Problem Statement

Background / Context

Vision-language models (VLMs) often use hundreds or thousands of visual tokens to represent an image, creating substantial computation and memory overhead during language-model inference. Many visual-token compression methods use attention scores to estimate which tokens are important for the current image or question. ApET takes a different approach: it is training-free and attention-independent, using linear reconstruction error to estimate visual-token redundancy. Tokens that are difficult to reconstruct from a small set of basis tokens are treated as more informative, while lower-ranked tokens are compressed through similarity-based merging [1].

ApET reports strong performance even under aggressive token reduction. This raises an important question: reconstruction error measures how distinctive a token is relative to other visual tokens, but distinctiveness is not necessarily the same as relevance to a particular question or contribution to the final answer. Related work also suggests that visual-token usefulness changes with task requirements and model depth [2], while prompt-agnostic compression may preserve information needed by later or different questions [3].

Problem Definition

This project will investigate why and when ApET's reconstruction-based token compression preserves task-relevant visual information, and where its assumptions break down. The study will focus on three questions:

1. How does reconstruction-based visual redundancy relate to task-relevant answer contribution, and how does this relationship change across task types and model layers?
2. How much of ApET's effectiveness comes from reconstruction-error-based token selection, and how much comes from similarity-based token merging?
3. Can the resulting analysis support a lightweight improvement to ApET, such as adaptive compression strength or an alternative reconstruction basis?

The goal is not to reproduce the complete ApET paper, but to use ApET as a case study for understanding the relationship between visual redundancy, question relevance, and answer preservation, and to explore whether this understanding can guide a more adaptive compression strategy.

4. Proposed Approach

Idea / Methodology

This project will use LLaVA-1.5-7B [4] with the public ApET implementation as the primary experimental system. The first step will be a limited reproduction of ApET to verify its basic accuracy-compression behavior at several visual-token budgets. Only the parts needed for the later analysis will be reproduced rather than all models, datasets, and baselines in the original paper. Minor implementation updates will be made where necessary for current software environments and GPU execution.

The main analysis will study the relationship between reconstruction-based redundancy and task-relevant information. For selected examples and model layers, the project will record ApET reconstruction errors and compare them with an intervention-based answer-contribution proxy inspired by the Information Horizon work [2]. The analysis will compare different task types and selected model depths. Same-image/different-question cases will also be used as controlled examples to study whether one question-independent compressed image representation preserves information needed by different questions.

A second analysis will study both whether token merging is beneficial and how merge targets should be chosen. At the same retained-token budget, reconstruction-error-based selection will first be evaluated with direct token removal versus ApET's nearest-retained-token similarity merge. The merge rule will then be examined using simple controls such as random merge-target assignment and an importance-protected strategy that prevents the highest-ranked

retained tokens from being modified. These experiments will test whether ApET's performance comes mainly from token selection, from recovering information through merging, from the choice of merge target, or from a combination of these factors.

Finally, the project will explore one lightweight extension motivated by the analysis. The main direction will be adaptive compression strength. Instead of assigning the same token budget to every example, statistics of the ApET reconstruction-error distribution will be used to estimate how aggressively an input can be compressed. If appropriate, a small classifier or regressor will be trained to select among several token budgets. An alternative reconstruction basis such as PCA/SVD will be considered as a secondary experiment if time permits.

Scope & Feasibility

The scope is intentionally limited to one primary VLM, LLaVA-1.5-7B, and a compact benchmark suite with distinct roles. GQA will support controlled diagnostic analysis, while MMStar and TextVQA will provide more recent and visually demanding evaluation settings. Large-scale reproduction of every model, dataset, and baseline in the ApET paper is outside the scope of this project. Basic reproduction and inexpensive ablations can use several hundred examples, while expensive token-level interventions will use smaller diagnostic subsets and only a few representative model layers.

Feasibility has already been partially validated through a preliminary test: LLaVA-1.5-7B with ApET successfully runs for both uncompressed and compressed visual-token inference on the available GPU resources. The main remaining computational cost comes from repeated analysis experiments rather than basic model execution. This cost can be controlled by limiting the number of examples, token interventions, layers, and compression settings. The proposed adaptive extension is lightweight and does not require training or fine-tuning the full VLM.

The target is to finish reproduction and analysis infrastructure in October, complete the main analysis and lightweight extension during November, and use December mainly for final evaluation, report writing, and presentation preparation.

5. Datasets and Resources

Datasets

GQA [5] will serve as a controlled base case for general and compositional visual question answering. Its multiple questions per image are useful for same-image/different-question analysis and detailed token-level interventions. A manageable subset will be used rather than the full dataset.

MMStar [7] will be the main modern broad benchmark. It contains 1,500 multiple-choice examples spanning six core capabilities and is designed so that visual input is necessary for answering the questions. It will test whether aggressive token compression preserves performance on a newer vision-dependent benchmark. Because the uncompressed LLaVA-1.5-7B model may itself have limited accuracy on MMStar, evaluation will report both overall accuracy and answer preservation on examples that the uncompressed model answers correctly.

TextVQA [6] will provide a complementary fine-detail and text-recognition setting, where compression may remove small but important visual evidence. A sampled subset will be used for this targeted analysis. MMBench [8] may be added as an additional broad modern benchmark, and OCRBench [9] as a stronger OCR-focused test, if time and resources permit.

Resources

The software stack will use Python, PyTorch, Hugging Face Transformers, and the public ApET/LLaVA implementation, together with standard numerical and statistical libraries. GPU development and small experiments will primarily use the available Kaggle 2×T4 environment (about 30 GPU hours per week). A Google Cloud budget of approximately \$50 is reserved for larger or more controlled final runs, potentially on a single L4 GPU. Local CPU resources will be used for dataset preparation, scoring, plotting, and unit tests.

6. Plan for Evaluation

Evaluation Metrics

Evaluation will cover both answer quality and the internal behavior of visual-token compression. For reproduction and compression experiments, the primary task metric will be the standard accuracy metric associated with each benchmark. In addition, an answer preservation rate will measure how often a compressed model preserves a correct answer from the uncompressed model: $APR(K) = P(\text{compressed correct} \mid \text{uncompressed correct})$, where K is the retained visual-token budget. This is especially important on newer challenge benchmarks such as MMStar, where

the base model may already miss many examples, because it separates compression-induced failures from the model's original errors.

For token-level analysis, ApET reconstruction error will be compared with an intervention-based answer-contribution proxy. Rank correlation (for example, Spearman correlation) and comparisons between low- and high-reconstruction-error token groups will be used to study whether visually distinctive tokens are also important to the model's answer. These results will be compared across task types and selected model layers. Selection-versus-merging experiments will compare answer accuracy and answer preservation under matched token budgets. An adaptive extension will be evaluated using both answer quality and the average number of retained visual tokens. Where useful, prefill latency, total inference latency, and peak GPU memory will also be measured under the same hardware and software configuration.

Baseline Comparisons

The main baseline will be uncompressed LLaVA-1.5-7B, which provides the reference answer quality and inference cost. Fixed-budget ApET will provide the main compression baseline. Random token selection at the same retained-token budget will provide a simple selection baseline.

For the merging study, controlled baselines will include: (1) direct dropping with no merge; (2) ApET's nearest-retained-token merge; (3) random merge-target assignment, to test whether similarity itself matters; and (4) an importance-protected merge in which the highest-ranked retained tokens are excluded as merge targets, to test whether modifying highly distinctive tokens is harmful. These merge variants will use the same retained-token budget and, where possible, the same selected-token set. For adaptive compression, fixed-budget ApET at a comparable average token count will be the direct baseline. An attention-based compression method may be included only if it can be integrated without substantially increasing the project scope.

Success Criteria

The main success criterion is not necessarily to outperform ApET with a new compression algorithm. The project will be considered successful if the experiments produce reproducible and interpretable evidence addressing the main research questions: how reconstruction error relates to answer contribution, how this relationship changes with task requirements and model depth, and how selection and merging contribute to ApET's behavior.

The initial reproduction should recover the broad behavior of ApET: stronger compression should reduce computational cost while eventually causing some loss in answer quality. Exact numerical reproduction of the original paper is not required because the evaluation scope and hardware will differ. For the adaptive extension, a positive result would be a better accuracy-token tradeoff than fixed-budget ApET, such as similar answer quality with fewer retained tokens on average. A well-controlled negative result will also be useful if it reveals limitations of reconstruction-error-based adaptation or merging.

7. Timeline & Milestones

Time	Milestone
Early-Mid Oct.	Reproduction and infrastructure: prepare datasets, verify baseline and ApET inference, update implementation as needed, and test several compression settings.
Mid-Late Oct.	Analysis instrumentation: log reconstruction errors, answer outputs, token selections, merge assignments, and intervention-based contribution measurements; finalize the evaluation protocol.
Early-Mid Nov.	Main analysis: run task/layer experiments, same-image/different-question cases, and selection/merging/merge-target ablations.
Mid-Late Nov.	Lightweight extension: test adaptive compression; if time permits, add a small PCA/SVD reconstruction or another optional control.
December	Freeze the setup, run only essential verification, prepare figures and tables, write the final report, and prepare the presentation.

8. References

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